

Temperature and kinetic energy

Learning outcomes

By the end of this section you should be able to:

- Interpret observable changes in physical properties and temperature during changes of state.
- Convert between values in the Celsius and Kelvin scales.

Changes of state are all around us in our daily lives but none of them have the possibility of affecting us as greatly as the melting of the polar ice caps. This is caused by global warming, a process that affects global temperatures. What sorts of activities or processes can you think of that contribute to global warming?

According to the National Oceanic and Atmospheric Administration, the past 30 years has seen a 95% decrease in the thickest ice in the Arctic 

(<https://www.washingtonpost.com/energy-environment/2018/12/11/arctic-is-even-worse-shape-than-you-realize/>). Here are ten things 

([https://www.un.org/actnow?](https://www.un.org/actnow?gclid=Cj0KCQjwwfiaBhC7ARIsAGvcPe77n3Y9PfN8GnPqmLypBrOyb7VfkElOWtzcZgzSHX6Dfc3sLa9H67kaAkck)

[gclid=Cj0KCQjwwfiaBhC7ARIsAGvcPe77n3Y9PfN8GnPqmLypBrOyb7VfkElOWtzcZgzSHX6Dfc3sLa9H67kaAkck](https://www.un.org/actnow?gclid=Cj0KCQjwwfiaBhC7ARIsAGvcPe77n3Y9PfN8GnPqmLypBrOyb7VfkElOWtzcZgzSHX6Dfc3sLa9H67kaAkck) you can do to help reduce global warming, according to the UN.

Changes of state occur when a substance loses or gains energy. When a substance loses or gains energy, it is usually accompanied by a change in temperature. An exception to this is during a change of state (or phase change) during which the temperature remains constant. This will be discussed in more detail later in the section.

Temperature scales

The two temperature scales used in the DP chemistry course are the Celsius scale and the Kelvin scale.

- The Celsius temperature scale is based on the freezing point of water (0 °C) and the boiling point of water (100 °C).
- The Kelvin scale is an absolute temperature scale where the lowest possible value is 0 K, known as absolute zero. At absolute zero, 0 K, particles have zero kinetic energy. The temperature in kelvin is directly proportional to the average kinetic energy of the particles in the substance. So the particles in a substance at a higher temperature have a higher average kinetic energy than those at a lower temperature.

Making connections

Do all particles in a substance have the same kinetic energy? In subtopic R2.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43127/>) you will learn about the distribution of kinetic energy

among the particles in a sample in more detail. These distributions are known as Maxwell-Boltzmann distributions.

Figure 1 shows a comparison of the two scales. Note that absolute zero (0 K) is equal to -273°C . The freezing point of water occurs at 0°C or 273 K and the boiling point of water is 100°C or 373 K. Unlike the Celsius scale, the Kelvin scale does not have negative temperatures. One kelvin is equal to one degree on the Celsius scale. For example, the difference between the boiling point and freezing point of water is 100°C (0°C to 100°C) on the Celsius scale which is the same as 100 K (273 K to 373 K) on the Kelvin scale.

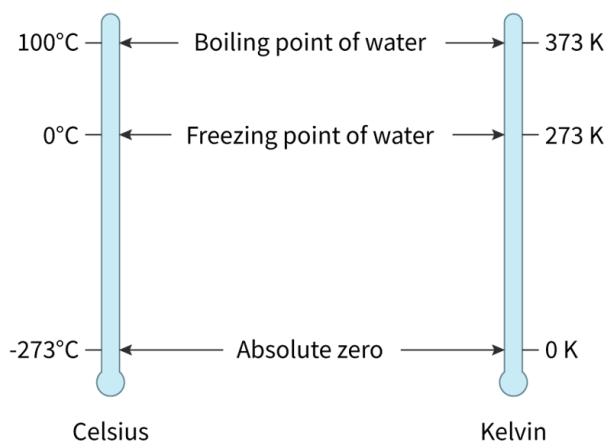


Figure 1. A comparison of the Kelvin and Celsius temperature scales.

 More information for figure 1

The image is a diagram comparing the Kelvin and Celsius temperature scales using two side-by-side thermometer shapes. On the left, the thermometer is labeled 'Celsius,' and on the right, it is labeled 'Kelvin.' Both thermometers have three labeled points.

1. At the bottom, labeled 'Absolute Zero':
2. Celsius scale reads -273 degrees.
3. Kelvin scale reads 0 degrees.
4. In the middle, labeled 'Freezing Point of Water':
5. Celsius scale reads 0 degrees.
6. Kelvin scale reads 273 degrees.
7. At the top, labeled 'Boiling Point of Water':
8. Celsius scale reads 100 degrees.
9. Kelvin scale reads 373 degrees.

This diagram illustrates that absolute zero is the same point on both scales but represented as -273°C on the Celsius scale and 0 K on the Kelvin scale. Similarly, the freezing and boiling points of water align across the scales but differ numerically, demonstrating the direct conversion between them by adding or subtracting 273.

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To convert between the two scales you either add or subtract 273 depending on which scale you are converting from. If you have a Celsius temperature and wish to convert to the Kelvin scale, you add 273. For example, 25 °C is 298 K ($25 + 273 = 298$). If you have a kelvin temperature and wish to convert to Celsius, you subtract 273. For example, 386 K is 113 °C ($386 - 273 = 113$).

Test your understanding by matching each of the temperatures in **Interactive 1**.

Drag the correct temperatures into the gaps to complete these sentences.

20 °C when converted to Kelvin is K

50 °C when converted to Kelvin is K

125 K when converted to degrees Celsius is °C

500 K when converted to degrees Celsius is °C

☒ Check

Interactive 1. Celsius—Kelvin Temperature Conversion.

Interactive 1. Match the temperatures.

International Mindedness

Throughout the DP chemistry course, students are encouraged to use SI units. SI stands for the International System of Units which is a system used globally that has seven base units. The SI unit of temperature is the kelvin (K). The remaining six SI units are the metre (m), kilogram (kg), second (s), mole (mol), ampere (A) and candela (cd). Be sure to look out for these during your studies.

Heating and cooling curves

Changes of state can be graphed by producing a heating (or cooling) curve. A heating curve shows how the state of matter changes as heat is added. **Figure 2** shows the heating curve for water. Note that a cooling curve would be the opposite, starting at a gas and ending with a solid with the temperature decreasing.

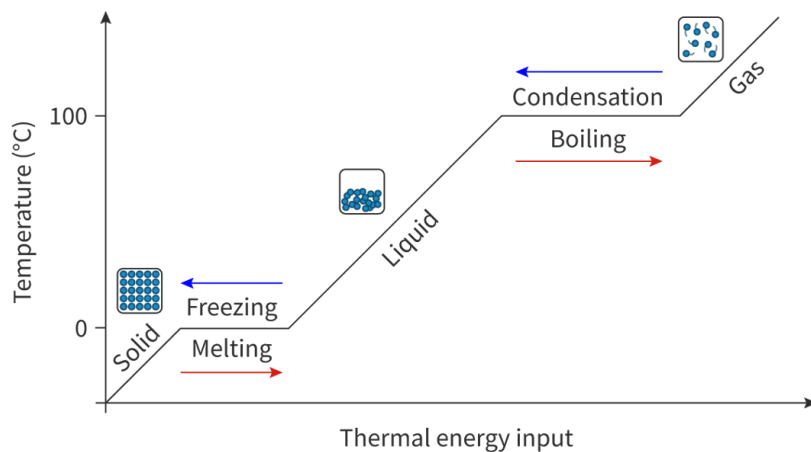


Figure 2. The heating curve for water.

 More information for figure 2

The image is a diagram depicting a graph of the heating curve for water. The X-axis is labeled "Thermal energy input," and the Y-axis is labeled "Temperature (°C)." The curve shows three main states of matter: solid, liquid, and gas. Solid is at the bottom, liquid is in the middle, and gas is at the top. Arrows indicate the transitions between these states: solid to liquid by melting, liquid to gas by boiling, liquid to solid by freezing, and gas to liquid by condensation. Each transition is marked along the curve. The temperature remains constant during phase changes (melting and boiling).

[Generated by AI]

Starting with solid water (ice), we can see that as heat is added, the temperature starts to increase. Once the ice reaches 0 °C, it starts to melt. During the melting process the temperature remains constant, in this case, 0 °C. The temperature remains constant during melting because all the added heat is being used to overcome the intermolecular forces that act between the particles. Once the ice is completely melted, the temperature starts to increase again. At 100 °C, the liquid water boils. Once again, we see that the temperature remains constant despite heat being added. This occurs because the added heat is being used to overcome the intermolecular forces that act between the water molecules. Once the liquid water has boiled it becomes a gas, which is known as steam.

Changes of state and heat

Finally we will classify each change of state as either absorbing heat or releasing heat. Melting, evaporation (or boiling) and sublimation are changes of state where heat is absorbed. Think of boiling water for example, to make it boil you need to add heat usually by heating it in a saucepan or in an electric kettle. Condensation, freezing and deposition are changes of state where heat is released. For example, when a gas condenses or water freezes, heat is released. **Table 1** has a summary of the changes of state together with the changes in heat.

Table 1. Summary of the changes of state.

Change of state	Description	Is heat absorbed or released?
Sublimation	Solid to gas with no liquid state	Absorbed
Deposition	Gas to solid with no liquid state	Released
Evaporation	Liquid to gas	Absorbed
Boiling	Liquid to gas	Absorbed
Condensation	Gas to liquid	Released
Freezing	Liquid to solid	Released
Melting	Solid to liquid	Absorbed

Use the activity in the next section (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/activity-temperature-and-kinetic-energy-id-43166/>) to investigate the relationship between temperature and kinetic energy.